

Working with Signed Rational Numbers

Adding and subtracting signed numbers, multiplying and dividing them, ordering rationals written in different forms, and evaluating mixed expressions in the right order

Before you start. These problems are deliberately mixed, so read each one before you reach for a rule.

- **Adding or subtracting?** Rewrite a subtraction as adding the opposite, then think about which number is further from zero.
- **Multiplying or dividing?** Count the negative signs: an even number of them gives a positive result, an odd number gives a negative one.
- **Several operations?** Brackets, then exponents, then multiplication and division *left to right*, then addition and subtraction left to right.
- **Comparing?** Put both numbers in the same form (decimals are usually easiest), and remember that further left on the number line means *smaller*.

1. Evaluate: $-8 + 3$

Answer: _____

2. Evaluate: $-6 - (-11)$

Rewrite the subtraction as an addition first, then finish.

Answer: _____

3. Evaluate: $(-4)(7)$

Answer: _____

4. Order these numbers from **least to greatest**:

$$-1, \quad -\frac{3}{4}, \quad -0.6, \quad \frac{1}{2}$$

Answer: _____

5. Evaluate: $-3 + 4(-2)$

Which operation has to happen first, and why?

Answer: _____

6. Evaluate and write the answer in lowest terms:

$$\left(-\frac{3}{4}\right) \left(\frac{8}{9}\right)$$

Answer: _____

7. Evaluate: $2.5 + (-7.25)$

Answer: _____

8. Write $<$, $>$ or $=$ between these two numbers, and say how you decided:

$$-\frac{5}{6} \quad \text{_____} \quad -0.8$$

Answer: _____

9. Evaluate: $(-2)^3 + 12 \div (-4)$

Show the value of each piece before you combine them.

Answer: _____

10. Priya evaluated $-24 \div (-6) \cdot (-2)$ and wrote -2 . Find her mistake, then evaluate the expression correctly.

Answer: _____

11. Evaluate: $-3(2 - 5) - \frac{4^2}{8}$
Write out each stage of your work.

Answer: _____

12. Challenge. Order these four numbers from **greatest to least**, then explain in one sentence how you handled the mix of decimals and fractions.

$$-2.5, \quad -\frac{11}{4}, \quad -2.05, \quad -\frac{9}{4}$$

Answer: _____